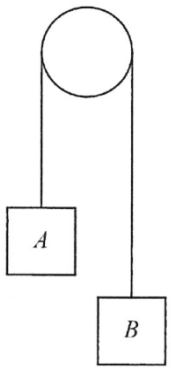
Forces: Pulleys

Questions 1–18

Question 1. The following questions use $g = 9.8$ **ms**$^{-2}$ Two particles A and B of mass m_A and m_B are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest.



The system is released from rest. In each case find the tension in the string and the acceleration of the system:

a) $m_A = 9$ kg, $m_B = 3$ kg
b) $m_A = 2$ kg, $m_B = 8$ kg
c) $m_A = 200$ g, $m_B = 500$ g

Question 2. Two particles A and B of masses $5m$ kg and $3m$ kg respectively are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. Find

a) the acceleration of the system.
b) the tension in the string in terms of m
c) the magnitude of the force exerted on the pulley while A and B are in motion.

Question 3. The following questions use exact values of g Two particles A and B of masses m kg and M kg respectively, where $M > m$, are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. The tension in the string is T N and the acceleration is a ms^{-2}. Show that

$$a = \left(\frac{M-m}{M+m}\right) g$$

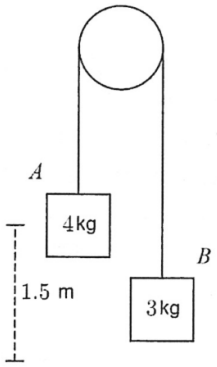
Question 4. Two particles A and B of masses km kg and m kg respectively where $k > 1$. They are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. The acceleration of the system is a ms^{-2}. Show that

$$a = \left(\frac{k-1}{k+1}\right) g$$

Question 5. The following questions use $g = 9.81$ **ms**$^{-2}$ Two particles A and B are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The masses hang with the string taut, and the system is released from rest. A has mass 2 kg and descends 0.8 metres in the first 4 seconds after the system is released. Find

a) the acceleration of the system.
b) the tension in the string.
c) the mass of B.

Question 6. Two particles A and B of masses 4 kg and 3 kg respectively are attached to the end of a light inextensible string which passes over a small smooth fixed pulley.

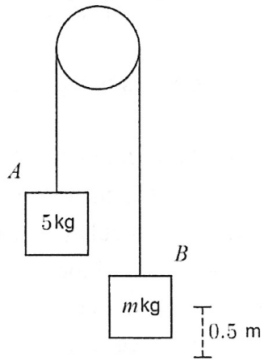


The masses hang with the string taut, and the system is released from rest. Initially, particle A begins 1.5 metres above the ground.
Find

a) the acceleration of the system.
b) the tension in the string.
c) the time it takes for A to hit the ground.
d) the greatest height of B above the floor, stating any modelling assumptions you have made.

Question 7. The following questions use $g = 9.8$ ms^{-2}

Two boxes A and B of masses 5 kg and m kg respectively are attached to the end of a light inextensible string which passes over a smooth small fixed pulley. The masses hang with the string taut, and the system is released from rest.



Box B starts 0.5 metres above the ground.
Box A ascends 0.2 metres in the first 0.5 seconds after the system is released.

a) Find the acceleration of the system.
b) Find the tension in the string.
c) Show that the mass of B is approximately 7 kg.
d) Find the velocity at which B hits the ground.

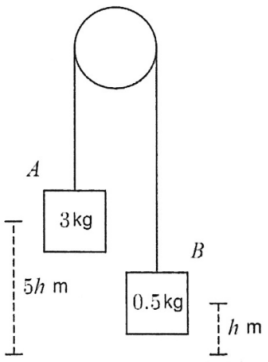
Question 7 (continued). Once Box B hits the ground, the string holding up box A becomes slack.

e) Given box A starts 1 metre above the ground, find the greatest height that box A reaches above the ground.
f) Find the time box A spends with the string not taut.

A different box (box C) is attached to the string, in replacement of box B. The system is re-set so that Box C starts 0.5 m above the ground.
Explain what box A will experience with its acceleration (including the direction) and the tension in the string in each of the following scenarios:

g) Box C has a larger mass than box B.
h) Box C has the same mass as box A.
i) Box C has a lower mass than box A.

Question 8. Two boxes A and B of masses 3 kg and 0.5 kg respectively are attached to the end of a light inextensible string which passes over a small smooth fixed pulley. The system is released from rest. Box A begins $5h$ metres from the ground and box B begins h metres from the ground.

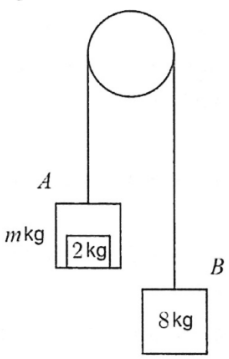


Find

a) the acceleration of the system.
b) the tension in the string.
c) Show that the time after release that both boxes are at the same height is given by

$$t = \sqrt{\frac{4h}{7}}$$

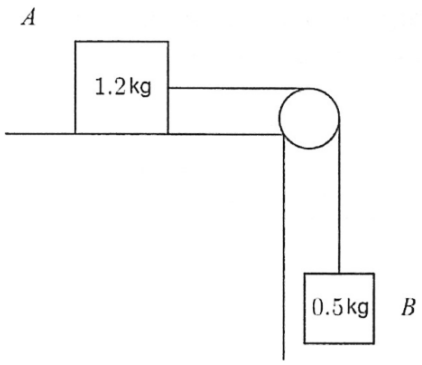
Question 9. A scale pan A of mass m kg hangs freely and is attached by a light inextensible string to a mass B which is 8 kg. The string passes over a small smooth fixed pulley. The masses hang with the string taut.



An object of mass 2 kg is placed into the scale pan and the system is released from rest. The scale pan A descends but doesn't exceed 1.96 ms^{-2}. Find

a) the range of the values of T where T N is the tension in the string.
b) the range of values of m

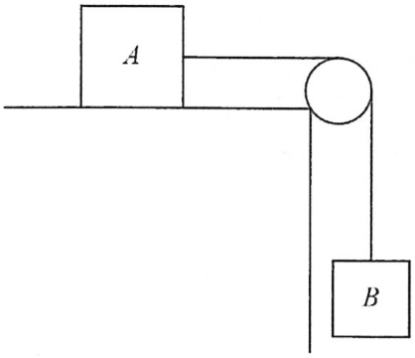
Question 10. Two particles, A and B have masses 1.2 kg and 0.5 kg respectively. They are connected by a light, inextensible string. Particle A lies on a smooth horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.



Find

a) the acceleration of the system.
b) the tension in the string.

Question 11. Two particles, A and B have masses m_A and m_B respectively. They are connected by a light, inextensible string. Particle A lies on a rough horizontal table and the string passes over a small round pulley which is fixed at the edge of a table. Particle B hangs freely. The resistance force acting on particle A is F N.



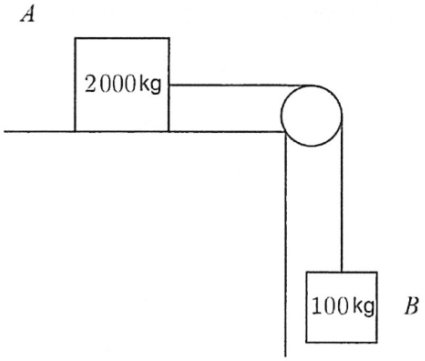
The system is released from rest.
Find, in each case, the acceleration of the system and the tension in the string.

a) $m_A = 2$ kg, $m_B = 3$ kg, $F = 20$
b) $m_A = 5$ kg, $m_B = 7$ kg, $F = 12$
c) $m_A = 200$ g, $m_B = 500$ g, $F = 2$
d) $m_A = 8$ kg, $m_B = 6$ kg, $F = 58.8$

Question 12. For each of the scenarios in question 11), find the time it takes for particle B to travel 1 m, assuming particle B begins over 1 metre from the floor.

Question 13. The following questions use $g = 10$ ms^{-2}

A van A of mass 2000 kg is attached with a cable to a pallet of building materials B which has mass 100 kg. The cable passes over a smooth fixed pulley, and B hangs freely.



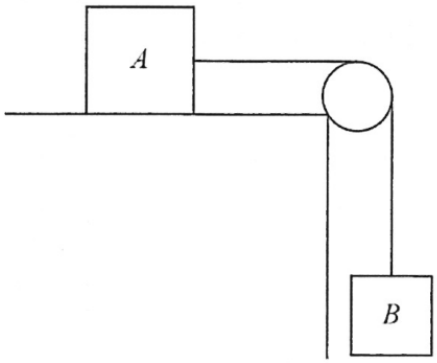
Initially the system is at rest and the cable is taut. The van begins to drive away from the pulley with driving force 2500 N. The resistance force acting on the van is 450 N.

a) Find the acceleration of the system.
b) Find the tension in the cable.
c) State two modelling assumptions that you have made.

The van drives forward for 10 seconds before the pallet is getting close to reaching the pulley.

d) Find the height that the pallet has been lifted.
e) State any assumptions in you have used.

Question 14. Two particles, A and B have masses M kg and m kg respectively. They are connected by a light, inextensible string. Particle A lies on a smooth horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.

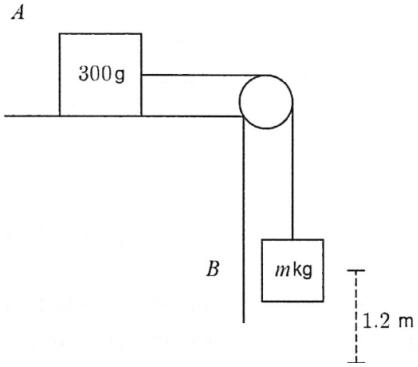


Show that

$$a = \frac{10m}{m + M}$$

Question 15. The following questions use $g = 9.8$ ms^{-2}

A wooden block A with mass 300 g sits on a rough table, attached to a string which passes over a small, smooth pulley which is fixed to the edge of the table. A plastic block B of mass m kg hangs freely at the end of the string.



The system is released from rest and block B descends. The resistance to motion on block A is 5 N.

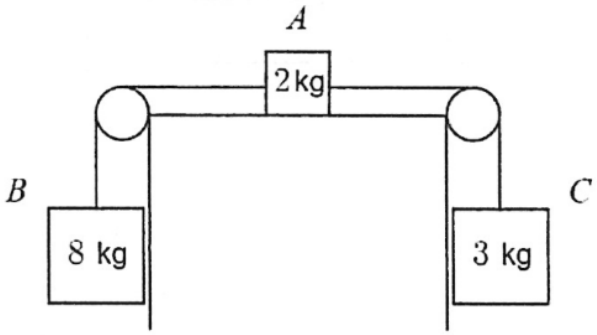
a) Find the acceleration of the system in terms of m

The block B initially starts 1.2 metres above the ground. It takes 2 seconds for B to reach the ground.

b) Find the value of m

c) State any modelling assumptions you have made.

Question 16. Particles A, B and C, have masses 2 kg, 8 kg and 3 kg respectively. A and B are attached to the ends of a light inextensible string. B and C are attached to the ends of another light inextensible string. B rests on a smooth table. A and C hang freely. Both strings pass over small smooth fixed pulleys at the edge of the table with the string taut.

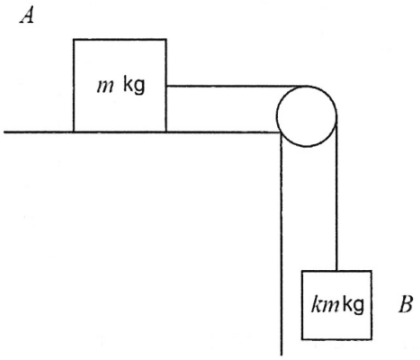


The system is released from rest. Find

a) the acceleration of the system.
b) the tension in the two strings.

Question 17. The following questions use exact values of g

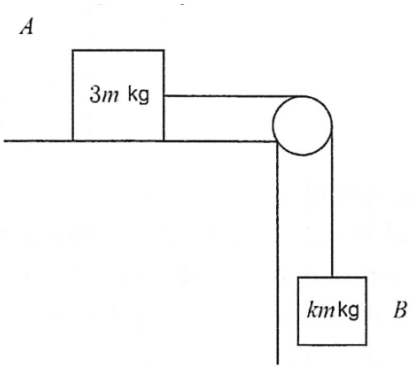
Two particles, A and B have masses m kg and km kg respectively where $k > 1$. They are connected by a light, inextensible string. Particle A lies on a rough horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.



The system is released from rest.
The resistance to motion on particle A is given as $\frac{1}{2}kmg$ N. Show that

$$a = \frac{kg}{2(1+k)}$$

Question 18. Two particles, A and B have masses $3m$ kg and km kg respectively where $k > 0$. They are connected by a light, inextensible string. Particle A lies on a rough horizontal table and the string passes over a small round smooth pulley which is fixed at the edge of a table. Particle B hangs freely.



A horizontal force of $20mg$ N is applied to A away from the pulley. The resistance to motion on particle A is given as $10mg$ N. Initially at rest, particle A travels 0.5 m in 10 s.

a) Find the acceleration of the system

b) Show that

$$k = \frac{1000g - 3}{100g + 1}$$